

MAT 322 HW08

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Problem 1. Let U be an open set in $\mathbb{R}^k$ and $f: U \rightarrow \mathbb{R}$ be a C^∞ function. Show that the graph of f is a k -dimensional C^∞ submanifold of $\mathbb{R}^{k+1}$.

Solution. Consider the projection map $\pi: \mathbb{R}^{k+1} \rightarrow \mathbb{R}^k$ given by $\pi(x_1, \dots, x_k, x_{k+1}) = (x_1, \dots, x_k)$. The derivative of this map is the $k \times (k+1)$ matrix

$$(\mathrm{d}\pi)_x = \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix}$$

so we can clearly see that π is C^∞ and has rank k everywhere. Letting $\Gamma = \{(x, f(x)) \in \mathbb{R}^k \times \mathbb{R} : x \in U\}$ be the graph of f , we see that if $p \in \Gamma$, then $f(U)$ is a neighborhood of p such that $\pi(f(U)) = U$ is open in $\mathbb{R}^k$; noting the properties of π mentioned above, this is enough to verify that Γ is a k -dimensional C^∞ manifold. $\square$

Problem 2. Let $c > a > 0$ be real. Consider the C^∞ function $f: \{(x, y, z) \in \mathbb{R}^3 : (x, y) \neq 0\} \rightarrow \mathbb{R}$ defined by

$$f(x, y, z) = (c - \sqrt{x^2 + y^2})^2 + z^2.$$

Show that $f^{-1}(a^2)$ is a C^∞ submanifold of $\mathbb{R}^3$ and describe it.

Solution. We can easily verify that f is a C^∞ function via some tedious computation.

Clearly $f^{-1}(a^2) = g^{-1}(0)$ where $g(x, y, z) = f(x, y, z) - a^2$, so we can say that $f^{-1}(a^2)$ is a C^∞ submanifold of $\mathbb{R}^n$ if $(\mathrm{d}g)_x$ is surjective for all $x \in f^{-1}(0)$, or equivalently $(\mathrm{d}f)_x$ is surjective for all $x \in f^{-1}(a^2)$. We have

$$(\mathrm{d}f)_{(x,y,z)} = \left(2(c - \sqrt{x^2 + y^2}) \cdot \frac{-2x}{2\sqrt{x^2 + y^2}}, 2(c - \sqrt{x^2 + y^2}) \cdot \frac{-2y}{2\sqrt{x^2 + y^2}}, 2z \right),$$

and to show that this is a surjective linear map $\mathbb{R}^3 \rightarrow \mathbb{R}$, we just need to verify that the components are not all zero when $f(x, y, z) = a^2$; for this, we check some cases.

- If $z \neq 0$ then obviously $2z \neq 0$ as well, so $(\mathrm{d}f)$ has rank 1 as desired.
- Otherwise suppose $z = 0$. Then $(c - \sqrt{x^2 + y^2})^2 = a^2$ implies that $\sqrt{x^2 + y^2}$ is either equal to $c - a$ or $c + a$. Since $c + a > c - a > 0$, $x^2 + y^2$ is nonzero, which implies that at least one of x and y is nonzero, say WLOG $x \neq 0$. In this case the corresponding component of $(\mathrm{d}f)$

$$2(c - \sqrt{x^2 + y^2}) \cdot \frac{-x}{\sqrt{x^2 + y^2}}$$

is nonzero (noting that $\sqrt{x^2 + y^2}$ is not equal to c because $a \neq 0$) so again the derivative has rank 1.

Therefore we can conclude that $f^{-1}(a)$ is a C^∞ submanifold of $\mathbb{R}^3$ with dimension $3 - 1 = 2$.

This set is a torus obtained by rotating a circle γ about the origin, where γ lies in the xz -plane such that its center is a distance of c from the origin and its radius is a . To verify this, one can see that f is invariant under rotation about the z -axis because such rotations preserve the distance $\sqrt{x^2 + y^2}$ to the origin, and then check that the cross section in the xz -plane is indeed the described circle $(c - x)^2 + z^2 = a^2$ (well, really it's the two circles $(c - |x|)^2 + z^2 = a^2$, but we can restrict to $x > 0$ because the other circle $(c + x)^2 + z^2 = a^2$ is the one opposite this circle on the torus). $\square$

Problem 3. Let $f: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be the function

$$f(x, y, z, w) = (x^2 - y^2 + z^2 - w^2, 2xy + 2zw).$$

Show that $f^{-1}(a, b)$ is a 2-dimensional C^∞ submanifold of $\mathbb{R}^4$ if $(a, b) \neq (0, 0)$.

Solution. Clearly f is C^∞ , because its components are polynomials which are C^∞ . To show that the pre-image of a nonzero point is a 2-dimensional C^∞ submanifold of $\mathbb{R}^4$ for each $x \in f^{-1}(a, b)$, we need to check that $(df)_{(x,y,z,w)}$ has rank 2 for each $(x, y, z, w) \in f^{-1}(a, b)$. We have

$$(df)_{(x,y,z,w)} = \begin{pmatrix} 2x & -2y & 2z & -2w \\ 2y & 2x & 2w & 2z \end{pmatrix}$$

If $(x, y) \neq (0, 0)$ then the left two rows of the matrix are independent, and if $(z, w) \neq (0, 0)$ then the right two rows are independent, so either of these forces $(df)_{(x,y,z,w)}$ to be surjective. In particular the only way the linear map isn't surjective is if $(0, 0, 0, 0) \in f^{-1}(a, b)$, but then this means $(a, b) = f(0, 0, 0, 0) = (0, 0)$, which we have forbidden. Therefore the derivative of f is always a surjective map $\mathbb{R}^4 \rightarrow \mathbb{R}^2$, so $f^{-1}(a, b)$ is a C^∞ submanifold of $\mathbb{R}^4$ with dimension $4 - 2 = 2$ as desired. $\square$

Problem 4. Let $C \subseteq \mathbb{R}^2$ be the **nodal curve** defined by the equation $x^2 = y^2$. Is C a C^∞ submanifold of $\mathbb{R}^2$?

Solution. No. We show that no neighborhood of $(0, 0)$ in the subspace C is homeomorphic to $\mathbb{R}^n$ for any positive integer n (and of course C^∞ manifolds are also C^0 manifolds). Suppose otherwise: then, for some open $U \subseteq C$, there is a homeomorphism $f: U \rightarrow \mathbb{R}^n$. Since C is the union of the lines $y = x$ and $y = -x$, the set $C \setminus \{(0, 0)\}$ has four connected components: $\{(x, x) : x > 0\}$, $\{(x, x) : x < 0\}$, $\{(x, -x) : x > 0\}$, and $\{(x, -x) : x < 0\}$. Then, since connectedness and disconnectedness is preserved by homeomorphism, we get that $\mathbb{R}^n \setminus \{f(0)\}$ must also have four connected components. However, this is impossible:

- If $n = 1$, then deleting $f(0)$ from $\mathbb{R}$ gives two connected components, $(-\infty, f(0))$ and $(f(0), \infty)$.
- If $n \geq 2$, then deleting $f(0)$ does not affect the connectedness of $\mathbb{R}^n$.

This implies that C does not have a C^0 n -dimensional atlas for any n , so it is not a C^0 submanifold of $\mathbb{R}^2$, let alone a C^∞ one. $\square$